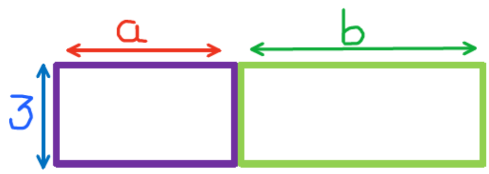
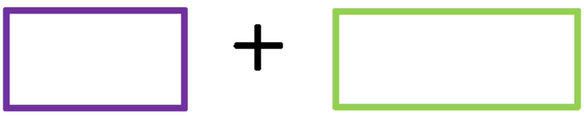
5H Factorisation expressions

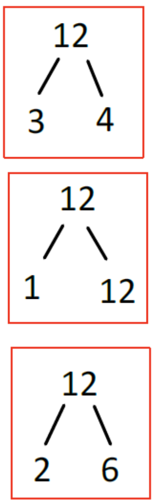
Learning Objective	To learn how to factorise algebraic expressions
Today We Are	To apply the distributive law to factorise algebraic expressions with numerical and simple algebraic highest common factors.
Indicators of Progress	• I can understand that factorising is the reverse procedure of expanding • I can find the highest common factor of two terms • I can factorise expressions

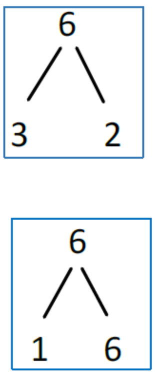
Lets Expand: $3(a+b)$





What are factors?





Key Words:

Integers:
Whole numbers, including positive & negative numbers, zero, and positive numbers.
Example: -3, 0, 7

Factors
Numbers that divide exactly into another number with no remainder.
Example: 2 and 4 are factors of 8.

Common Factor
A factor that two or more numbers share.
Example: 2 is a common factor of 6 and 8.

Highest Common Factor
The largest factor that two or more numbers share.
Example: The HCF of 12 and 18 is 6.

Expanding
Multiplying brackets to remove them and make an expression bigger.
Example: $3(x+2) = 3x + 6$

Factors of 12
{1, 2, 3, 4, 6, 12}

Factors of 6
{1, 2, 3, 6}

What are the **Common Factors** between 12 and 6? { 1, 2, 3, 6 }

What is the **Highest Common Factor** between 12 and 6? { 1, 2, 3, $\boxed{6}$ }

How about factorising algebraic expressions ?

- Reverse of **expanding.**
- When you have an expanded expression and identify common factors to turn the expression back into brackets

Expanding Brackets

V.S

Factorising Brackets

$6x(2x+3)$

Step 1:
- **Multiply** the term outside the brackets by all the terms inside the bracket separately.
- Remember Dinner party activity , the guest greeted all the hosts!

$6x(2x + 3)$

$(6x \times 2x) + (6x \times 3)$

Step 2:
- **Simplify** the expression

$= 12x^2 + 18x$

$4a + 8$

Step 1:
- Find the **Factors** of each of the numbers (integers)
- Identify the **Common Factors** and the **Highest Common Factor** between each number

Factors of 4 { 1, 2, 4 }

Factors of 8 { 1, 2, 4, 8 }

Common Factors: **{ 1, 2, 4}**

Highest Common Factor: **4**

Step 2:
- Identify any other common **variables/ pronumerals/ letters**
  - If there are no other commonalities between the two terms, proceed to **step 3.**

Step 3: Lets put our highest common Factor (and any common variables/letters) in front of the bracket.

$4a + 8$

$= 4(a + 2)$

Highest Common factor

1, 2, 4, 8

1, 2, 4, 8, 16

1, 2, 7, 14

1, 3, 7, 21

1, 3, 3, 9

1, 2, 3, 4, 6, 8,

$= 2(x+2)$

2 x   2 2

$= 3(x+3)$

3 x   3 3

$= 4(a-2)$

4 a   4 2

$= 5(a-3)$

5 a   5 3

$= 3c(3+4d)$

3c 3   3c 4

$= 2(6a-5)$

2 6a   2 5